

MCP-Powered Multi-Agent Air Quality Reasoning System

1. Introduction

Air pollution is one of the most pressing global challenges affecting environmental quality, respiratory health, and urban livability. While cities deploy monitoring stations, they often fail to integrate multi-dimensional signals such as vehicular emissions, visual pollution signatures, AQI trends, and health impact indicators.

This project proposes an **AI-Driven Air Pollution Risk Intelligence Agent**, powered by multiple deep learning models exposed via a **Model Context Protocol (MCP)** server. Each model functions as a distinct "tool," enabling modular reasoning, real-time prediction, and actionable risk advisory.

2. Problem Definition

Existing systems typically provide only raw pollution statistics (AQI values). They lack the capability to:

- **Forecast** pollution progression over time.
- **Identify** pollution sources, such as specific traffic behaviors.
- **Detect** visual pollution signatures from city imagery.
- **Map** pollution data directly to population health risks.

There is currently no holistic agent capable of learning pollution patterns, interpreting emissions, and generating contextual health advisories simultaneously.

3. Innovation Statement

This system introduces four autonomous deep learning models, each addressing a unique dimension of urban air pollution. The core innovation lies in decomposing pollution intelligence into machine-learned expertise layers rather than relying on a single monolithic model.

- **Temporal View:** Predicts future AQI levels using LSTM-based time-series forecasting.
- **Spatial View:** Uses a CNN to classify polluted city environments from image signals.
- **Source View:** Uses a mobility-informed DNN model to estimate emission levels from vehicle/traffic behavior.
- **Human Impact View:** Uses a health impact model to infer public risk based on correlated environmental and medical datasets.

MCP-Enabled AI Orchestration: All models are wrapped as MCP tools, allowing higher-level agents to combine predictions, reason over multi-modal signals, and generate actionable guidance.

4. Social Relevance

This work directly contributes to the UN Sustainable Development Goals (**SDG 3:** Good Health, **SDG 11:** Sustainable Cities, **SDG 13:** Climate Action).

Key Contributions:

- **Cleaner Urban Planning:** Data-driven insights for city infrastructure.
- **Smart City Mitigation:** Real-time triggers for pollution control.
- **Policy Regulation:** Evidence-based traffic emission regulation.
- **Public Awareness:** Early warnings for vulnerable groups and general citizen health.

Governments and NGOs can utilize this system for real-time risk dashboards, targeted public health advisories, and emission-based routing recommendations.

5. Datasets Used

Component	Dataset Name	Source & Link	Purpose
LSTM Forecasting	Air Quality Data in India	Kaggle Link	Train LSTM to predict AQI classes (Good, Satisfactory, Moderate, Poor, Very Poor).
CNN Visual Classifier	Air Pollution Image Dataset (India/Nepal)	Kaggle Link	Train CNN to detect if urban scenes exhibit visual pollution/haze.
Emission Predictor	Vehicle Emission Dataset	Kaggle Link	Learn how vehicle type, fuel, road condition, and traffic affect pollution levels.
Health Advisor	Air Quality & Health Impact Dataset	Kaggle Link	Train a model to map pollution data to public health risk outcomes.

6. Deep Learning Models & Justification

6.1 LSTM Time-Series Classifier (AQI Forecasting)

- **Why Best Suited:** AQI is highly dependent on past temporal patterns. LSTMs model multivariate sequences better than traditional statistical models and can learn daily, weekly, and seasonal patterns.
- **Function:** Predicts the next-state AQI category.

6.2 Transfer Learning CNN (EfficientNet-B0 / ResNet50)

- **Why Best Suited:** Pollution cues (smog, haze) are visual and subtle. CNNs capture global scene representation, and transfer learning is effective for smaller domain-specific datasets.
- **Function:** Classifies imagery as "Polluted" or "Non-Polluted."

6.3 DNN with Embeddings (Emission Prediction)

- **Why Best Suited:** The dataset contains mixed categorical and numeric features. Deep embeddings allow for learning vehicle type, fuel, and traffic context while capturing non-linear interactions.
- **Function:** Classifies vehicle emission levels (Low / Medium / High).

6.4 Multi-Task DNN (Health Impact Estimation)

- **Why Best Suited:** Health impact is derived from multiple interacting features. Learning the score and class jointly improves representation and supports both quantitative and qualitative advisory generation.
- **Function:** Outputs a Health Risk Score (0–100) and a Risk Class (Very Low to Very High).

9. Innovation Breakdown

- **Multi-dimensional Pollution Intelligence:** Unlike typical AQI dashboards, this system integrates temporal forecasting, perception-based pollution, source attribution, and health advisory generation.
- **Multi-Model & Multi-Agent AI:** Each model is served modularly via MCP, allowing for composable reasoning.
- **Real-World Social Impact:** Applicable to smart cities, environmental control boards, healthcare monitoring, and disaster resilience.
- **Interpretability & Interoperability:** Models can be upgraded independently without redesigning the entire system architecture.